

Spiders and scorpions

Spiders and scorpions belong to an ancient family of predatory animals called chelicerates (“kell-ISS-er-ates”), all of which have special mouthparts that they use either as pincers or fangs. Modern chelicerates are small, but their earliest ancestors grew to gigantic sizes and were among the top predators of their time. The biggest of these prehistoric monsters were the sea scorpions.

Pterygotus

terry-GOAT-us



- **When** 400–380 million years ago (Late Silurian to Middle Devonian)
- **Fossil location** Europe, N. America
- **Habitat** Shallow seas
- **Length** Up to 7 ft 4 in (2.3 m)

Pterygotus was a sea scorpion that grew larger than a fully grown man. Using its colossal eyes, it scanned the water for prey such as fish and trilobites. Perhaps it hid half-buried in sand until victims wandered close by, before lashing its tail to produce a violent burst of speed and snatching up the animal in its claws. Fossils have been found worldwide, and some experts think *Pterygotus* not only terrorized the seas but swam up rivers and into lakes as well.



FAMILY FACT FILE

Key features

- Segmented bodies and jointed limbs
- Hard external skeleton (exoskeleton)
- Pincerlike feeding claws or fangs
- Four pairs of walking legs

When

The chelicerates appeared late in the Ordovician Period, about 445 million years ago. Over 77,000 identified species exist today.



DID YOU KNOW...?

Animals with jointed legs and external skeletons (such as insects, spiders, and scorpions) are called arthropods. Sea scorpions were the largest arthropods that ever lived—giant versions of the ones we find today in the yard. Today, arthropods are small but in the distant past they grew to greater sizes, perhaps because the Earth's air contained more oxygen, making it easier for arthropods to breathe and grow.